CptS 484: Software Requirements

WRS Evolution

Requirements Elicitation

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Revision History

Date	Version	Changes	Editor
10/10/2026	1.0	Added Introduction and Section 2	Runlin Zhao
10/12/2026	1.1	Added 3.1 problem, goals, and domain	Keagen Chapman
10/13/2026	1.2	Added 3.2 requirements, traceability tables, process records	Nathaniel Gaines
10/18/2026	1.3	Editing changes and references to external documents	Keagen Chapman
10/18/2026	1.4	Updated 3.1 so the problems, goals and objectives match 3.2 and Section 7; added 3.2.3 specifications and the repo link	Nathaniel Gaines

1. Introduction

1.1 Purpose

The objective of the project is to develop a smartphone application that helps blind people navigate indoors. Moving around inside a large building is difficult without sight. Room numbers, signs and floor directories are visual, and GPS, which most navigation apps rely on, does not work well inside buildings. A white cane or a guide dog helps the user avoid what is directly in front of them, but it cannot tell them where a room is or which hallway leads to the elevator.

Existing tools do not fully solve this problem. Google Indoor Maps shows floor plans of large buildings but is designed for sighted users, and Microsoft Soundscape used 3D audio to guide blind users but focused on outdoor travel. Our application, Akita, is named after Helen Keller's dog, a symbol of loyal guidance for people who cannot see.

Akita accepts a destination by voice, finds the user's location using wireless beacons placed at each room, plans a route, and guides the user with spoken step counts, turn cues and vibration. It warns the user about obstacles on the route and places an emergency call if the user falls. Figure 1 shows how Akita works with the user and the devices around it.

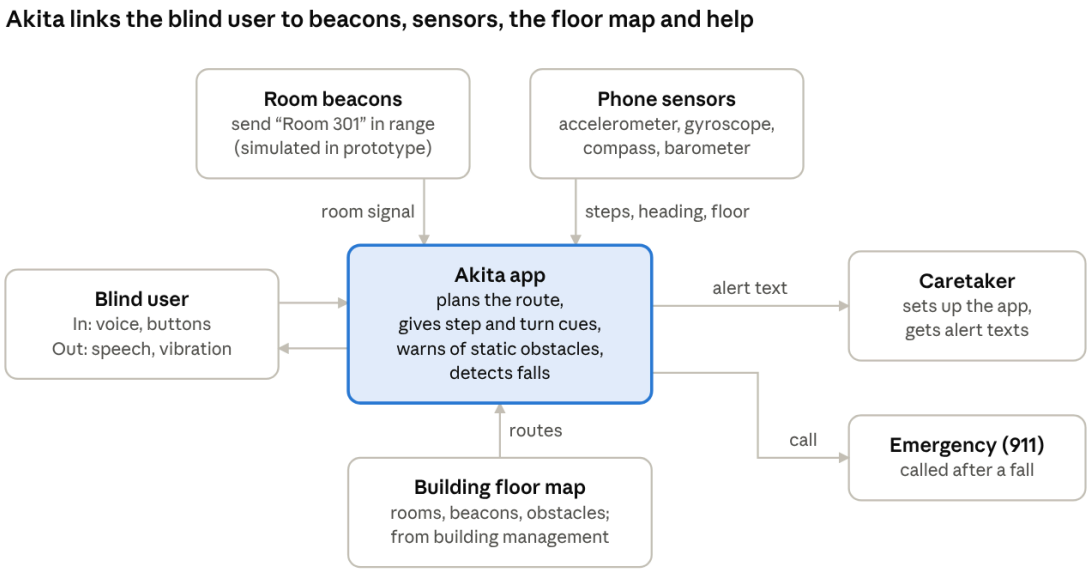


Figure 1: Akita system context

Figure 1: Akita system context. The app combines beacon signals, phone sensors and the building floor map to guide the user, and contacts the caretaker or 911 when help is needed.

1.2 Scope

Akita is used inside buildings that may have several floors, each with classrooms, offices, restrooms, lounges, elevators and stairs. It guides the user from one room or location to another, not inside a room, and it does not cover outdoor travel. Each room and significant location is assumed to have a wireless beacon that tells the phone where the user is; in the prototype these beacon signals are simulated. Obstacles are limited to static ones such as pillars, trash cans, corners and walls. Akita supplements the user's cane or guide dog and does not replace them.

For Phase I, the team demonstrates the requirements through scenarios and a prototype mock-up rather than a finished app. The core functionalities of the application are:

- Accept a destination from the user by voice and confirm it
- Find the user's current location and plan a route to the destination
- Guide the user with step counts, turn cues and floor changes
- Warn the user about static obstacles on the route
- Place emergency calls and messages after a fall

1.3 Objectives and Success Criteria

The project objectives and success criteria are that the software must:

- accept a destination by voice without the user looking at the screen
- know which room the user is in from beacon signals
- plan a safe route and let the user choose between options
- give clear walking instructions in steps, personalized to the user
- tell the user when to stop, turn, and change floors
- warn the user about obstacles before reaching them
- call for help quickly when the user falls
- be fully usable by a blind person
- be customizable to each user

- be easy to extend to new buildings and features

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
Akita	The indoor navigation app developed by Team Luminous.
Beacon	A small wireless transmitter at a room or location that tells the phone where the user is.
Caretaker	A person, such as a family member, who helps the blind user set up the app and is contacted in an emergency.
Clock-face direction	A direction given as a clock hour, e.g., "turn to 3 o'clock" means turn right.
Floor map	A digital map of a building's floors, rooms, beacons and obstacles used to plan routes.
FR / NFR	Functional Requirement / Non-Functional Requirement.
Haptic feedback	Vibration patterns used as cues, e.g., to stop or turn.
Stride length	The length of one step, estimated from the user's height.
TalkBack / VoiceOver	The built-in screen readers of Android and iOS.
THEIA	The course project name for the indoor navigation problem.
WRS	World, Requirements, Specification.

1.5 Overview

In section 2, the issues with the preliminary definition given were laid out. It consists of the domain issues, functional requirements issues, and non-functional requirements issues. For each part, the preliminary items are listed first, followed by the issues found in them. Options and the rationale for each choice were made.

In section 3, there are two main topics, W and RS. The W section states the problems and the corresponding goals, together with the improved understanding of the domain, stakeholders, functional and non-functional objectives. The RS section lists the functional and non-functional requirements. Section 4 covers the preliminary prototype and user manual, and section 5 covers traceability.

2. Issues with Preliminary Definition Given

2.1 Issues with II.1 The Domain, Stakeholders, Functional and Non-Functional Objectives

PD_ID	Preliminary Domain Description
PD1	The domain is indoors, which can consist of multiple floors, each of which possibly hosts multiple classrooms, offices, bathrooms, lounges, elevators, etc.

PD_ID	Preliminary Domain Description
PD2	The primary stakeholder is a blind person who needs to navigate indoors.
PD3	Secondary stakeholders include a caretaker who sets the configuration of the app and comes to aid the blind person in case of emergency, staff in the accessibility department, and the police.
PD4	Functional objectives include navigating indoors from one location to another in the same or different buildings that are connected to each other.
PD5	Non-functional objectives include safe, fast, and comfortable navigation.
PD6	Usability is a top priority; since a blind person cannot read the screen, the app should use features such as voice recognition.
PD7	Use as many sensors as are available on an advanced smartphone.

2.1.1 DI1

Domain Issue ID	DI1
PD_ID	PD1: The domain is indoors, which can consist of multiple floors …
Description	Ambiguous. "Indoors" does not say which buildings are supported or how the app knows the layout of a building and where the user is.
Option 1	Support any building by mapping it automatically as the user walks.
Option 2	Support buildings that have a digital floor map and a beacon at each room and significant location.
Choice	Option 2
Rationale	Automatic mapping is not reliable enough to guide a blind person safely. A floor map with beacons gives predictable routes and lets the app know which room the user is in.
Satisfied by	Improved Domain (Section 3.1.3)

2.1.2 DI2

Domain Issue ID	DI2
PD_ID	PD2: The primary stakeholder is a blind person who needs to navigate indoors.

Domain Issue ID	DI2
Description	Ambiguous. Blindness ranges from no vision to low vision, and users may or may not use a cane or guide dog. The level of impairment changes how the app should interact with the user.
Option 1	Design only for users with no vision.
Option 2	Design for users with no vision who use a cane or guide dog, and add simple support for low-vision users (large buttons, high contrast).
Choice	Option 2
Rationale	Designing for users with no vision covers the hardest case, and low-vision support is easy to add. Assuming a cane or guide dog keeps Akita a supplement rather than the user's only protection.
Satisfied by	Stakeholders (Section 3.1.3)

2.1.3 DI3

Domain Issue ID	DI3
PD_ID	PD3: … a caretaker who sets the configuration of the app and comes to aid the blind person in case of emergency …
Description	1. Ambiguous. It is unclear what the caretaker can configure and whether they can see the user's location at all times. 2. Conflict. Constant tracking helps safety but takes away the user's privacy and independence.
Option 1	The caretaker helps set up the app and receives the user's location only in an emergency.
Option 2	The caretaker can change settings remotely and see the user's location at all times.
Choice	Option 1
Rationale	The blind user is the primary stakeholder and should stay in control of their own data. The caretaker can still help when it matters most.
Satisfied by	Stakeholders (Section 3.1.3)

2.1.4 DI4

Domain Issue ID	DI4
PD_ID	PD4: … in the same or different buildings that are connected to each other.
Description	Inconsistent. Buildings may be connected by outdoor walkways, which conflicts with an indoor-only domain.

Domain Issue ID	DI4
Option 1	Include outdoor walkways between buildings.
Option 2	Include only indoor connections such as skybridges and tunnels.
Choice	Option 2
Rationale	Outdoor walkways would need GPS and street-level safety features, which are outside the project scope. Indoor connections keep the domain consistent.
Satisfied by	Improved Domain (Section 3.1.3)

2.1.5 DI5

Domain Issue ID	DI5
PD_ID	PD5: Non-functional objectives include safe, fast, and comfortable navigation.
Description	Conflict. The fastest route may use stairs or crowded areas, which is less safe or comfortable. No priority between the three objectives is given.
Option 1	Make safety a requirement for every route; among safe routes, use the user's preference for fastest or most comfortable.
Option 2	Always choose the fastest route.
Choice	Option 1
Rationale	An unsafe route is never acceptable for a blind user. Whether speed or comfort matters more is a personal choice, so the user decides.
Satisfied by	Improved Non-Functional Objectives (Section 3.1.3)

2.1.6 DI6

Domain Issue ID	DI6
PD_ID	PD6: … utilizing the voice recognition capability of smartphones.
Description	Incomplete. Voice recognition may fail in noisy hallways, and the user may not want to speak in quiet places. No other way to interact is mentioned.
Option 1	Use voice as the main input, with physical buttons and vibration as backups.
Option 2	Use voice input and spoken output only.
Choice	Option 1

Domain Issue ID	DI6
Rationale	Voice is the fastest way for a blind user to give commands, but buttons and vibration still work when voice fails.
Satisfied by	Improved Non-Functional Objectives (Section 3.1.3)

2.1.7 DI7

Domain Issue ID	DI7
PD_ID	PD7: Use as many sensors as are available on an advanced smartphone.
Description	Ambiguous. Phones have different sensors, and using all of them drains the battery. It is unclear which sensors are required.
Option 1	Use every sensor the phone has.
Option 2	Use a common set of sensors: Bluetooth for beacons, accelerometer and gyroscope for steps and falls, compass for direction, and barometer for floor changes.
Choice	Option 2
Rationale	These sensors are available on almost all smartphones and cover what the app needs, without draining the battery.
Satisfied by	Improved Domain (Section 3.1.3)

2.2 Issues with II.2 Software System Requirements: Functional Requirements

PFR_ID	Preliminary FR Description
PFR1	Accept from the user the destination location to go; possibly suggest or confirm a destination using the user's routine schedule or habit.
PFR2	Figure out the routes to reach the destination, inform the user of the options, and accept the user's preference.
PFR3	Tell the user to walk a distance (e.g., 2 minutes before turning, or walk for 30 steps).
PFR4	Tell the user to stop at the right place to turn.
PFR5	Detect obstacles and tell the user what to do to avoid collision.
PFR6	Place emergency calls and messages, possibly after detecting a fall or when the system has lost its current location.

PFR_ID	Preliminary FR Description
PFR7	Figure out the next action(s) based on the user's schedule or habit, and suggest or accept the user's choice.
PFR8	More that you can think of.

2.2.1 FRI1

FR Issue ID	FRI1
PFR_ID	PFR1: Accept from the user the destination location to go …
Description	1. Speech recognition may mishear the destination, sending the user to the wrong place. 2. It is unclear how a destination can be named (room number, office, "nearest restroom").
Option 1	Start navigating as soon as a destination is heard.
Option 2	Accept a room number, name or category, read the destination back to the user, and start only after the user confirms.
Choice	Option 2
Rationale	Reading the destination back catches recognition errors before the user starts walking, and accepting different kinds of names makes the app easier to use.
Satisfied by	FR (Section 3.2.1)

2.2.2 FRI2

FR Issue ID	FRI2
PFR_ID	PFR2: Figure out the routes to reach the destination, inform the user of the options …
Description	1. It is unclear how many route options to give. 2. A blind user cannot compare routes on the screen, and a long spoken list is hard to remember.
Option 1	Suggest the best route based on the user's preference, and read up to two other routes only if the user asks.
Option 2	Read all possible routes and let the user choose.
Choice	Option 1
Rationale	One suggested route keeps the interaction short, while the user can still ask for other options.
Satisfied by	FR (Section 3.2.1)

2.2.3 FRI3

FR Issue ID	FRI3
PFR_ID	PFR3: Tell the user to walk a distance (e.g., 2 minutes before turning, or walk for 30 steps).
Description	Inconsistent. The examples use two different units, minutes and steps. Walking speed and step length differ from person to person.
Option 1	Give distances in time (minutes or seconds).
Option 2	Give distances in steps, using a step length estimated from the user's height.
Option 3	Give distances in meters or feet.
Choice	Option 2
Rationale	Steps are easy for a blind user to count. Estimating step length from the user's height makes the count fit each person, while walking time changes with crowds and pace.
Satisfied by	FR (Section 3.2.1)

2.2.4 FRI4

FR Issue ID	FRI4
PFR_ID	PFR4: Tell the user to stop at the right place to turn.
Description	Ambiguous. It is unclear how early the user is warned and how the direction of the turn is given.
Option 1	Warn the user a few steps before the turn, give a stop cue by voice and vibration at the turn, and give the direction as left/right or a clock-face direction.
Option 2	Say "turn left" or "turn right" only at the turning point.
Choice	Option 1
Rationale	An early warning gives the user time to slow down, vibration works in noisy places, and clock-face directions describe turns that are not exactly left or right.
Satisfied by	FR (Section 3.2.1)

2.2.5 FRI5

FR Issue ID	FRI5
PFR_ID	PFR5: Detect obstacles and tell the user what to do to avoid collision.

FR Issue ID	FRI5
Description	1. It is unclear which obstacles are included (furniture, people, open doors). 2. Detecting moving obstacles in real time requires the camera to point forward, which is hard while holding a cane.
Option 1	Detect all obstacles, including moving ones, with the camera.
Option 2	Record static obstacles (pillars, trash cans, corners) on the floor map and warn the user before reaching them.
Choice	Option 2
Rationale	Static obstacles can be warned about reliably without the camera, and moving obstacles are outside the project scope. The user's cane or guide dog still handles anything unexpected.
Satisfied by	FR (Section 3.2.1)

2.2.6 FRI6

FR Issue ID	FRI6
PFR_ID	PFR6: Place emergency calls and messages, possibly after detecting a fall or when the system has lost its current location.
Description	1. It is unclear who should be contacted. 2. A dropped phone could be mistaken for a fall and cause a false alarm. 3. Losing location is treated like a fall, but a lost user usually needs directions, not an ambulance.
Option 1	Call 911 immediately when a fall is detected or the location is lost.
Option 2	After a fall, ask "Are you OK?"; if there is no answer within 30 seconds, call 911 and text the caretaker the user's location. If the location is lost, help the user find a nearby room first, then offer to call the caretaker.
Choice	Option 2
Rationale	Asking first avoids false alarms while still getting help to a user who cannot respond. Treating a lost location differently sends the right kind of help.
Satisfied by	FR (Section 3.2.1)

2.2.7 FRI7

FR Issue ID	FRI7
PFR_ID	PFR7: Figure out the next action(s) based on the user's schedule or habit …

FR Issue ID	FRI7
Description	1. Redundant. This overlaps with the schedule-based suggestion in PFR1. 2. It is unclear where the schedule comes from.
Option 1	Merge PFR7 into PFR1 and use a schedule the user enters or imports from the phone's calendar.
Option 2	Learn the user's habits automatically from past trips.
Choice	Option 1
Rationale	Merging removes the duplicate. A user-entered schedule is predictable and does not require storing the user's location history.
Satisfied by	FR (Section 3.2.1)

2.2.8 FRI8

FR Issue ID	FRI8
PFR_ID	PFR8: More that you can think of.
Description	Incomplete. Some functions a blind user needs are missing, such as changing floors and recovering from a wrong turn.
Option 1	Add guidance for elevators and stairs, rerouting when the user goes off route, repeating the last instruction, and telling the user where they are.
Option 2	Add no further functions.
Choice	Option 1
Rationale	Without these functions, the user could get stuck in the middle of a route, especially between floors.
Satisfied by	FR (Section 3.2.1)

2.3 Issues with II.3 Software System Requirements: Non-Functional Requirements

PNFR_ID	Preliminary NFR Description
PNFR1	The system shall help the user safely navigate indoors.
PNFR2	The system shall lead the user through the fastest route.
PNFR3	The system shall lead the user through the route that the user would feel the most comfortable with.
PNFR4	The system shall be usable for blind people.
PNFR5	The system shall be ubiquitous.

PNFR_ID	Preliminary NFR Description
PNFR6	The system shall be customizable to every user: e.g., volume, the interval of instructions, etc.
PNFR7	The system shall be easily extensible to accommodate variations in interface, language, definitive needs of the user, new features, new sensors and hardware, etc.

2.3.1 NFRI1

NFR Issue ID	NFRI1
PNFR_ID	PNFR1: The system shall help the user safely navigate indoors.
Description	Ambiguous. What is considered safe navigation? As written, it cannot be tested.
Option 1	Never route the user through a hazard they want to avoid (such as stairs), and warn about obstacles at least 3 steps before reaching them.
Option 2	Keep "safe" as a general goal.
Choice	Option 1
Rationale	Clear criteria make safety measurable and testable. The 3-step distance gives the user enough time to react.
Satisfied by	NFR (Section 3.2.2)

2.3.2 NFRI2

NFR Issue ID	NFRI2
PNFR_ID	PNFR2: The system shall lead the user through the fastest route.
Description	1. Ambiguous. Fastest could mean the shortest distance or the shortest time. 2. Conflict. The fastest route may not be the safest one.
Option 1	Choose the shortest distance.
Option 2	Choose the shortest time based on the user's own walking pace, among safe routes only.
Choice	Option 2
Rationale	The user experiences time, not distance, and blind users often walk slower than average. Choosing only among safe routes resolves the conflict with PNFR1.
Satisfied by	NFR (Section 3.2.2)

2.3.3 NFRI3

NFR Issue ID	NFRI3
PNFR_ID	PNFR3: The system shall lead the user through the route that the user would feel the most comfortable with.
Description	Ambiguous. Comfort is different for each person, and the app cannot know it unless the user tells it.
Option 1	Use a fixed rule, such as the route with the fewest turns.
Option 2	Let the user set comfort preferences, such as avoiding stairs, fewer turns, or quieter hallways.
Choice	Option 2
Rationale	Only the user knows what feels comfortable. Saved preferences also make the requirement testable.
Satisfied by	NFR (Section 3.2.2)

2.3.4 NFRI4

NFR Issue ID	NFRI4
PNFR_ID	PNFR4: The system shall be usable for blind people.
Description	Ambiguous. What makes the app usable for a blind person? No way to measure it is given.
Option 1	Every function works with VoiceOver/TalkBack and voice commands, and each main task takes no more than 3 commands or button presses.
Option 2	Follow general accessibility guidelines.
Choice	Option 1
Rationale	Concrete criteria can be checked against the prototype, while general guidelines cannot show the app is actually usable.
Satisfied by	NFR (Section 3.2.2)

2.3.5 NFRI5

NFR Issue ID	NFRI5
PNFR_ID	PNFR5: The system shall be ubiquitous.
Description	Ambiguous. Ubiquitous could mean working in every building, on every phone, or at all times. Working in every building conflicts with DI1, which limits the app to buildings with a floor map and beacons.

NFR Issue ID	NFRI5
Option 1	Work in any building.
Option 2	Work on both iOS and Android, in every supported building, and without internet once the building map is downloaded.
Choice	Option 2
Rationale	This meaning is achievable and consistent with DI1. Working without internet matters because signal can be weak inside buildings.
Satisfied by	NFR (Section 3.2.2)

2.3.6 NFRI6

NFR Issue ID	NFRI6
PNFR_ID	PNFR6: The system shall be customizable to every user: e.g., volume, the interval of instructions, etc.
Description	Incomplete. The list ends with "etc.", and it is unclear who changes the settings and how a blind user can do it without seeing a menu.
Option 1	Let the user change settings by voice: volume, speech rate, instruction interval, distance unit, vibration, route preference, and emergency contacts. The caretaker can help with the first setup.
Option 2	Let only the caretaker change volume and instruction interval.
Choice	Option 1
Rationale	A complete list makes the requirement testable, and voice settings let the user stay independent.
Satisfied by	NFR (Section 3.2.2)

2.3.7 NFRI7

NFR Issue ID	NFRI7
PNFR_ID	PNFR7: The system shall be easily extensible …
Description	Ambiguous. "Easily extensible" has no measure, and "definitive needs of the user" is not defined.
Option 1	Treat extensibility as a general design goal.
Option 2	Keep voice input, audio output, location, maps and sensors in separate parts of the app, and keep all spoken text in language files, so new buildings, languages and sensors can be added without changing the rest.

NFR Issue ID	NFRI7
Choice	Option 2
Rationale	Separating the parts of the app makes each change small and easy to check. Language files make it simple to add languages later.
Satisfied by	NFR (Section 3.2.2)

[3] WRS

3.1. W

3.1.1. Problem

Problem ID	Problem Description	Corresponding Goals
P1	How can blind people navigate building interiors and avoid obstacles?	G1, G2, G3
P2	How can blind people navigate exteriors? (out of scope except building entrances, see DI4)	G3
P3	How to identify obstacles.	G1, G2
P4	How to communicate with the user non-visually.	G4
P5	The app must be available on various mobile devices.	G5, G7
P6	What information should be communicated to the user.	G2
P7	A blind user who is lost or injured indoors cannot easily get help.	G6
P8	Users differ (speech rate, step length, language, caretaker needs), so one fixed behavior does not fit everyone.	G7

3.1.2. Goals

Goal ID	Goal Description	Backward Traceability	Forward Traceability
G1	Use the phone's camera and sensors to observe the area around the user.	P1, P3	IFRO1, INFRO2
G2	Identify obstacles and tell the user what matters about them (direction, distance, how to avoid). A pre-trained model such as YOLO may be used in the extended demo.	P1, P3, P6	IFRO1, IFRO2, INFRO2, INFRO4
G3	Use indoor positioning (Bluetooth beacons) and a building floor map to navigate between indoor locations; GPS is used only at building entrances.	P1, P2	IFRO3, INFRO4
G4	Receive spoken commands and give information through audio and vibration.	P4	IFRO2, INFRO3
G5	The app is available on iOS and Android and can be extended to other platforms.	P5	INFRO5

G6	Let the user summon help and let the caretaker know where the user is.	P7	IFRO4, INFRO1
G7	Let the user and caretaker tailor the app to the user.	P5, P8	IFRO5, INFRO5

3.1.3. Improved understanding of Domain, Stakeholders, Functional, and Non-Functional Objectives

3.1.3.1. Improved Domain

Improved Domain ID	Improved Domain Description
ID1	Akita operates inside supported buildings: multi-floor buildings with classrooms, offices, restrooms, lounges, elevators and stairs, and indoor connections (such as skybridges and tunnels) to other buildings. Each supported building has a digital floor map and a Bluetooth beacon at each room and significant location. Akita uses a common set of phone sensors: Bluetooth, accelerometer, gyroscope, compass and barometer. Only static obstacles recorded on the floor map are considered. Outdoor navigation is out of scope (DI1, DI4, DI7). development tool.

3.1.3.2. Stakeholders

- Primary Stakeholders
    - Team Luminous: Keagen Chapman, Nathaniel Gaines, Runlin Zhao
    - Blind Users
- Secondary stakeholders
    - University: Bolong Zeng, Washington State University
    - The families of blind users
    - Caretakers (such as family members), who help set up the app and are contacted in an emergency
    - Accessibility staff, who provide floor maps and beacon data for each building
    - Emergency services (911), reached only through the emergency call

3.1.3.3. Improved Functional Objectives

Improved FR Objective ID	Objective Description	Alleviates Problems	Achieves Goals
IFRO1	The system will warn the user of static obstacles on the route, with direction and distance.	P1, P3, P6	G1, G2
IFRO2	The system will give spoken and vibration instructions and accept spoken commands.	P4, P6	G2, G4

IFRO3	The system will use the building floor map and beacon-based location to guide the user to destinations.	P1, P2	G3
IFRO4	The system will send an emergency call or message with the user's last known location when help is needed.	P7	G6
IFRO5	The system will let the user and caretaker configure destinations, preferences and contacts.	P8	G7

3.1.3.4. Improved Non-Functional Objectives

Improved NFR Objective ID	Objective Description	Alleviates Problem	Achieves Goal
INFRO1	The system will keep location and contact data secure and share it only in an emergency or with consent.	P7	G6
INFRO2	Navigation will be safe: hazard warnings are timely and accurate.	P1, P3	G1, G2
INFRO3	The system will be fully usable without sight.	P4	G4
INFRO4	Routes will be fast and comfortable.	P1, P6	G2, G3
INFRO5	The system will be customizable, extensible and available on common phones.	P5, P8	G5, G7

3.2. RS

3.2.1. Functional Requirements

FR ID	Description
FR1	the system shall accept a destination as a room number, name or category (for example, "nearest restroom"), read the destination back to the user, and start navigation only after the user confirms.
Satisfies Functional Requirement Issue	FRI1
Satisfies Objectives	IFRO2
Satisfied by prototype feature	Voice Destination Entry

FR ID	Description
FR2	the system shall suggest the next destination from a schedule that the user enters or imports from the phone's calendar,

	and shall start navigation only after the user confirms.
Satisfies Functional Requirement Issue	FRI7
Satisfies Objectives	IFRO5
Satisfied by prototype feature	Destination Suggestions

FR ID	Description
FR3	the system shall plan routes on the building floor map using only indoor connections between buildings, announce the best route for the user's preference first, and read up to two other routes only if the user asks.
Satisfies Functional Requirement Issue	FRI2, DI1, DI4
Satisfies Objectives	IFRO3
Satisfied by prototype feature	Route Planning

FR ID	Description
FR4	the system shall find the user's current room or location from the beacon signals of the building and update it as the user moves, using the accelerometer, gyroscope, compass and barometer to follow steps, direction and floor changes.
Satisfies Functional Requirement Issue	DI1, DI7
Satisfies Objectives	IFRO3
Satisfied by prototype feature	Location Tracking

FR ID	Description
FR5	the system shall give walking distances in steps, estimated from the user's height, or in the distance unit the user selects in settings.
Satisfies Functional Requirement Issue	FRI3
Satisfies Objectives	IFRO2, IFRO3
Satisfied by prototype feature	Step Instructions

FR ID	Description
FR6	the system shall warn the user a few steps before each turn, give a stop cue by voice and vibration at the turning point, and state the turn as left or right or as a clock-face direction.
Satisfies Functional Requirement	FRI4, DI6

Issue	
Satisfies Objectives	IFRO2, IFRO3
Satisfied by prototype feature	Turn and Stop Cues

FR ID	Description
FR7	the system shall warn the user, before the user reaches them, of static obstacles recorded on the floor map (such as pillars, trash cans and corners).
Satisfies Functional Requirement Issue	FRI5
Satisfies Objectives	IFRO1
Satisfied by prototype feature	Obstacle Warnings

FR ID	Description
FR8	the system shall, after a detected fall, ask the user "Are you OK?" and, if there is no answer within 30 seconds, call 911 and text the caretaker the user's location; when the user's location is lost, it shall first help the user find a nearby room and then offer to call the caretaker.
Satisfies Functional Requirement Issue	FRI6, DI3
Satisfies Objectives	IFRO4
Satisfied by prototype feature	Emergency Help

FR ID	Description
FR9	the system shall repeat the last instruction and tell the user where they are when the user asks.
Satisfies Functional Requirement Issue	FRI8
Satisfies Objectives	IFRO2
Satisfied by prototype feature	Navigation Controls

FR ID	Description
FR10	the system shall detect when the user goes off the route and plan a new route from the current location.
Satisfies Functional Requirement Issue	FRI8
Satisfies Objectives	IFRO3
Satisfied by prototype feature	Automatic Rerouting

FR ID	Description
FR11	the system shall guide the user to and through elevators and stairs when the route changes floors and tell the user when the new floor is reached.
Satisfies Functional Requirement Issue	FRI8
Satisfies Objectives	IFRO3
Satisfied by prototype feature	Floor Change Guidance

FR ID	Description
FR12	the system shall let the user change volume, speech rate, instruction interval, distance unit, vibration, route preference (including comfort preferences such as avoiding stairs, fewer turns or quieter hallways) and emergency contacts by voice, and shall let the caretaker help with the first setup.
Satisfies Functional Requirement Issue	NFRI6, NFRI3, DI3
Satisfies Objectives	IFRO5
Satisfied by prototype feature	Settings and Caretaker Setup

3.2.2. Non-functional Requirements

NFR ID	Nonfunctional Requirement 1
NFR1	the system shall never route the user through a hazard the user wants to avoid (such as stairs) and shall warn the user about an obstacle at least 3 steps before the user reaches it.
Operationalized Functional Requirements	FR3, FR7
Satisfies Nonfunctional Requirement Issue	NFRI1, DI5
Satisfies Non-functional Objective	INFRO2
Constrains	P1, P3
Satisfied by prototype feature	Obstacle Warnings

NFR ID	Nonfunctional Requirement 2
NFR2	the system shall choose, among safe routes only, the route with the shortest estimated time based on the user's own walking pace.
Operationalized Functional Requirements	FR3, FR10

Satisfies Nonfunctional Requirement Issue	NFRI2, DI5
Satisfies Non-functional Objective	INFRO4
Constrains	P1
Satisfied by prototype feature	Route Planning

NFR ID	Nonfunctional Requirement 3
NFR3	the system shall choose a route that satisfies the comfort preferences the user has saved (for example, avoiding stairs, fewer turns, quieter hallways).
Operationalized Functional Requirements	FR3, FR12
Satisfies Nonfunctional Requirement Issue	NFRI3
Satisfies Non-functional Objective	INFRO4
Constrains	P1, P6
Satisfied by prototype feature	Route Planning

NFR ID	Nonfunctional Requirement 4
NFR4	the system shall be usable without sight: every function works with VoiceOver or TalkBack and voice commands, each main task takes no more than 3 commands or button presses, voice is the main input with buttons and vibration as backups, and buttons are large with high contrast for low-vision users.
Operationalized Functional Requirements	FR1, FR6, FR9, FR12
Satisfies Nonfunctional Requirement Issue	NFRI4, DI2, DI6
Satisfies Non-functional Objective	INFRO3
Constrains	P4
Satisfied by prototype feature	Navigation Controls

NFR ID	Nonfunctional Requirement 5
NFR5	the system shall run on iOS and Android in every supported building (one with a floor map and beacons) and keep navigating without internet once the building map is downloaded.
Operationalized Functional Requirements	FR3, FR4, FR5, FR6
Satisfies Nonfunctional Requirement Issue	NFRI5

Satisfies Non-functional Objective	INFRO5
Constrains	P5
Satisfied by prototype feature	Location Tracking

NFR ID	**Nonfunctional Requirement 6**
NFR6	the system shall let the user change the settings listed in FR12 by voice.
Operationalized Functional Requirements	FR12
Satisfies Nonfunctional Requirement Issue	NFRI6
Satisfies Non-functional Objective	INFRO5
Constrains	P8
Satisfied by prototype feature	Settings and Caretaker Setup

NFR ID	**Nonfunctional Requirement 7**
NFR7	the system shall keep voice input, audio output, location, maps and sensors in separate parts of the app and keep all spoken text in language files, so new buildings, languages and sensors can be added without changing the rest.
Operationalized Functional Requirements	FR1, FR4, FR7
Satisfies Nonfunctional Requirement Issue	NFRI7, DI1
Satisfies Non-functional Objective	INFRO5
Constrains	P5, P8
Satisfied by prototype feature	Settings and Caretaker Setup

NFR ID	**Nonfunctional Requirement 8**
NFR8	the system shall give the caretaker the user's location only in an emergency.
Operationalized Functional Requirements	FR8, FR12
Satisfies Nonfunctional Requirement Issue	DI3
Satisfies Non-functional Objective	INFRO1
Constrains	P7
Satisfied by prototype feature	Emergency Help

3.2.3. Specifications

Functional Specification ID	Functional Requirement
FS1	Input: spoken room number, name or category. The system listens up to 10 seconds, matches the phrase against the building's location list, and says "Go to <name>, correct?". Navigation starts only after a spoken "yes". "No" or no answer within 10 seconds restarts the prompt. After 2 failed matches the system offers the large-button entry instead.
Satisfies Functional Requirement	FR1
Satisfies Objectives	IFRO2, INFRO3
Satisfied by prototype feature	Voice Destination Entry

Functional Specification ID	Functional Requirement
FS2	Input: a schedule the user enters or imports from the phone's calendar. Before a scheduled event, the system suggests that event's room once ("Your next class is in room 205, go there?") and starts navigation only after a spoken "yes". A declined suggestion is not repeated for the same event.
Satisfies Functional Requirement	FR2
Satisfies Objectives	IFRO5
Satisfied by prototype feature	Destination Suggestions

Functional Specification ID	Functional Requirement
FS3	Input: confirmed destination and current location. Output within 3 seconds: the best route for the user's saved preference, with estimated time and floor changes. Up to two other routes are read only if the user asks. Routes use only indoor connections.
Satisfies Functional Requirement	FR3
Satisfies Objectives	IFRO3, INFRO4
Satisfied by prototype feature	Route Planning

Functional Specification ID	Functional Requirement
FS4	Input: beacon IDs and signal strength, plus accelerometer, gyroscope, compass and barometer readings. Output: current room or hallway segment, updated at least every 2 seconds while the user moves. If no beacon is received for 15 seconds, the location is "lost" and the lost-location flow in FS8 starts.
Satisfies Functional Requirement	FR4

Satisfies Objectives	IFRO3, INFRO5
Satisfied by prototype feature	Location Tracking

Functional Specification ID	Functional Requirement
FS5	Output: "Walk <n> steps", where steps = distance / step length and step length = 0.415 x the user's height (0.7 m if no height is set). For distances over 20 steps the count is repeated at the halfway point. The user can switch to meters or feet in settings.
Satisfies Functional Requirement	FR5
Satisfies Objectives	IFRO2, IFRO3
Satisfied by prototype feature	Step Instructions

Functional Specification ID	Functional Requirement
FS6	Output: a first warning 10 steps before each turn ("Turn left in 10 steps"), then at the turning point a stop cue by voice and a vibration pattern ("Stop. Turn left now"). Turns are given as left or right, or as a clock-face direction if the user chooses. At the destination: "You have arrived at <name>."
Satisfies Functional Requirement	FR6
Satisfies Objectives	IFRO2, IFRO3
Satisfied by prototype feature	Turn and Stop Cues

Functional Specification ID	Functional Requirement
FS7	Input: static obstacles recorded on the floor map. For each obstacle within 4 meters ahead on the route, the system says "<obstacle> ahead in <n> steps. Keep <left/right>" at least 3 steps before the user reaches it (NFR1). Obstacle warnings interrupt other speech and add a distinct vibration.
Satisfies Functional Requirement	FR7
Satisfies Objectives	IFRO1, INFRO2
Satisfied by prototype feature	Obstacle Warnings

Functional Specification ID	Functional Requirement
FS8	Trigger: a detected fall. The system asks "Are you OK?" within 5 seconds and waits 30 seconds for a spoken "I'm OK". With no answer it calls 911 and texts the caretaker the user's last known location. If the location is lost, it first helps the user find a nearby room, then offers to call the caretaker.
Satisfies Functional Requirement	FR8
Satisfies Objectives	IFRO4, INFRO1

Satisfied by prototype feature	Emergency Help

Functional Specification ID	Functional Requirement
FS9	Input: the spoken commands "repeat" and "where am I". "Repeat" replays the last instruction within 1 second. "Where am I" announces the current room or hallway segment, or the last known one if the location is lost.
Satisfies Functional Requirement	FR9
Satisfies Objectives	IFRO2, INFRO3
Satisfied by prototype feature	Navigation Controls

Functional Specification ID	Functional Requirement
FS10	Trigger: the user is more than 5 meters off the route for 5 seconds, or an obstacle blocks the path for 5 seconds. The system says "Rerouting" and plans a new route from the current location within 3 seconds, using the same rules as FS3.
Satisfies Functional Requirement	FR10
Satisfies Objectives	IFRO3, INFRO4
Satisfied by prototype feature	Automatic Rerouting

Functional Specification ID	Functional Requirement
FS11	When a route changes floors, the system says "Take the elevator (or stairs) to floor <n>" according to the user's preference, and guides the user to it. On reaching the beacon of the new floor it says "Floor <n> reached", followed by the next walking instruction (for example "Exit and turn left").
Satisfies Functional Requirement	FR11
Satisfies Objectives	IFRO3
Satisfied by prototype feature	Floor Change Guidance

Functional Specification ID	Functional Requirement
FS12	Settings the user can change by voice: volume (0-100%), speech rate (0.5x-2x), instruction interval (3-15 seconds), distance unit (steps, meters, feet), vibration on or off, route preference (fastest, fewest turns, avoid stairs, quieter hallways), and up to 5 emergency contacts. A caretaker can help with first setup using a pairing code.
Satisfies Functional Requirement	FR12
Satisfies Objectives	IFRO5, INFRO5
Satisfied by prototype feature	Settings and Caretaker Setup

[4-5] Preliminary Prototype, Prototype Interface Mock-ups, User Manual

See “Project Akita — Phase I Prototype and User Manual” external document

[7] Traceability

Problem	Goals	Objectives	FRs	NFRs
P1 Navigate interiors/obstacles	G1, G2, G3	IFRO1, IFRO3, INFRO2, INFRO4	FR3, FR4, FR5, FR6, FR7, FR10, FR11	NFR1, NFR2, NFR3
P2 Navigate exteriors (out of scope, DI4)	G3	IFRO3	FR3 (indoor connections only)	NFR5
P3 Identify obstacles	G1, G2	IFRO1, INFRO2	FR7	NFR1
P4 Communicate non-visually	G4	IFRO2, INFRO3	FR1, FR5, FR6, FR9	NFR4
P5 Multiple devices	G5, G7	INFRO5	FR4	NFR5, NFR7
P6 What info to give	G2	IFRO1, IFRO2, INFRO4	FR5, FR6, FR7	NFR3
P7 Lost/injured, no help	G6	IFRO4, INFRO1	FR8	NFR8
P8 Users differ	G7	IFRO5, INFRO5	FR2, FR12	NFR6, NFR7

[8] References

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[9] Appendix I: Process Details

9.1. Phase 1 Roles

Name	Role	Responsibilities
Runlin Zhao	Team leader and Communication Liaison	WRS Section 1 (Introduction) and Section 2 (Issues); formatting and combining the final WRS document; AI usage report; team repository and final submission; main contact with Dr. Zeng
Nathaniel Gaines	Meeting Coordinator; document owner for the requirements and plan	WRS Section 3.2 (functional and non-functional requirements, specifications) and the traceability matrix; alignment of Section 3.1 with Section 3.2; Phase I project plan (preliminary and final); meeting records; presentation slides 6-7 (Lesson 6 analysis); presentation video editing
Keagen Chapman	Technical Research Lead; prototype and presentation	WRS Section 3.1 (problems, goals, objectives); prototype mock-up and user manual; presentation slides 4, 5 and 8; reserving meeting rooms

9.2. Meetings

Date	Time	Attendees	Roles	Activities	Outputs
09/02/2026	3:30 PM, Spark 215	Keagen Chapman, Runlin Zhao, Nathaniel Gaines	Facilitator: Nathaniel Gaines; Scribe: Runlin Zhao	First meeting: reviewed the next submission and discussed a team name	Plan for the preliminary project plan
09/14/2026	3:30 PM, Spark 215	Keagen Chapman, Runlin Zhao, Nathaniel Gaines	Facilitator: Nathaniel Gaines; Scribe: Runlin Zhao	Worked on the preliminary project plan	Plan sections drafted for the 9/20 submission
10/05/2026	3:00 PM, Spark 215 (Nathaniel joined late by Discord voice)	Keagen Chapman, Runlin Zhao, Nathaniel Gaines	Facilitator: Nathaniel Gaines; Scribe: Runlin Zhao	Reviewed Phase I deliverables, split the WRS sections and presentation work, set	Phase I task plan; WRS document created

				deadlines	
10/13/2026	3:15 PM, Discord call	Keagen Chapman, Runlin Zhao, Nathaniel Gaines	Facilitator: Nathaniel Gaines; Scribe: Runlin Zhao	Reviewed WRS sections; planned the presentation and slides	Presentation slides shared; Section 2 revised; recording set for Friday
10/16/2026	3:00 PM, Zoom	Keagen Chapman, Runlin Zhao, Nathaniel Gaines	Facilitator: Runlin Zhao; Scribe: none	Recorded the presentation of the four AS-IS/TO-BE scenarios and the analysis	Presentation recording (later trimmed and re-edited)

9.3. Activities

Team Luminous formed on 9/1 and held its first meeting on 9/2. On 9/4 Runlin emailed Dr. Zeng to ask whether a functional app is required later in the course, and his reply that Phase I should show understanding of the requirements shaped the project scope. Between 9/9 and 9/20 the team wrote the preliminary project plan: Runlin drafted the overview (Section 1), Nathaniel wrote Sections 2 to 3.2, and the plan was submitted on 9/20. Dr. Zeng's feedback on 9/23 asked the team to remove template text and format Section 1.

From 9/21 to 10/6 the team reviewed the Preliminary Definition and identified issues. On 10/5 the team split the Phase I work and created the WRS document. Runlin wrote the Introduction and the issue analysis (10/10 to 10/13). Keagen wrote the problems, goals and objectives in Section 3.1 (10/12). Nathaniel wrote the requirements, specifications and traceability in Section 3.2 (10/13 to 10/18).

From 10/13 to 10/16 the team wrote the AS-IS/TO-BE scenarios and the Lesson 6 analysis of the top-priority scenario (scenario 3), and recorded the presentation on 10/16. Keagen wrote the prototype mock-up and user manual (10/17 to 10/18). On 10/18 Nathaniel edited the video, aligned Section 3.1 with the traceability tables and revised the project plan, and Runlin combined the sections into one WRS document, wrote the AI usage report and prepared the repository.